

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |          |          |           |          |          |
|----------|-----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (c) | (i)   | Either opposite charges <b>or</b> attractive force [on plates]                                                                                                                                                                                                                                                                                                                                                                              |                 | 1        |          | 1         |          |          |
|          |     | (ii)  | Either work done (1)<br>and conservation of energy applied (or implied) (1)<br><b>OR</b> $\text{energy} = \frac{1}{2} \frac{Q^2}{C}$ (1)<br>and $C$ decreases (1)<br><b>OR</b> $\frac{Q}{V} = \frac{\epsilon_0 A}{d}$ so $V \propto d$ (1)<br>$E = \frac{1}{2} QV$ hence correct (1)<br><b>OR</b> energy stored in field (1)<br>field has greater volume (1)<br><b>OR</b> increased the potential energy (1)<br>Due to attractive force (1) |                 | 2        |          | 2         |          |          |
|          |     | (iii) | 2 or 3 valid equations <b>used</b> e.g. $\frac{1}{2} \frac{Q^2}{C}$ and $C = \frac{\epsilon_0 A}{d}$ (1)<br>Convincing working leading to $\frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$ (if they quote the 2 equations above then simply writing the equation is acceptable) (1)                                                                                                                                                                 |                 | 2        |          | 2         | 2        |          |
|          |     | (iv)  | Work = force $\times$ distance quoted (1)<br>Hence $F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$ or distance = $d$ and Bethan correct<br><b>OR</b> Bethan wrong at large separations or when $d = 0$<br><b>OR</b> good conclusion well argued (1)                                                                                                                                                                                              |                 |          | 2        | 2         | 1        |          |
|          |     |       | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b>        | <b>8</b> | <b>4</b> | <b>13</b> | <b>8</b> | <b>0</b> |